§1 Fisher Information Matrix

1. Deriving the FIM from KL Divergence

We are given the KL Divergence:

$$K(w) = \int p(x|w^{(0)}) \cdot \log \left[\frac{p(x|w^{(0)})}{p(x|w)} \right] dx$$

Taking the first partial derivative with respect to w_j:

$$\frac{\partial K}{\partial w_j} = -\int p(x|w^{(0)}) \frac{\partial \log p(x|w)}{\partial w_j} dx$$

Taking the second partial derivative with respect to w_k:

$$\frac{\partial^2 K}{\partial w_j \partial w_k} = -\int p(x|w^{(0)}) \frac{\partial^2 \log p(x|w)}{\partial w_j \partial w_k} dx$$

Using the hint for the second derivative of the log probability:

$$\frac{\partial^2 \log p}{\partial w_j \partial w_k} = \frac{1}{p} \frac{\partial^2 p}{\partial w_j \partial w_k} - \left(\frac{\partial \log p}{\partial w_j} \right) \left(\frac{\partial \log p}{\partial w_k} \right)$$

Substituting this back into the integral:

$$\frac{\partial^2 K}{\partial w_j \partial w_k} = -\int p(x|w^{(0)}) \left[\frac{1}{p(x|w)} \frac{\partial^2 p(x|w)}{\partial w_j \partial w_k} - \frac{\partial \log p(x|w)}{\partial w_j} \frac{\partial \log p(x|w)}{\partial w_k} \right] dx$$

Evaluating this at the true parameter $w = w^{(0)}$, the $p(x|w)$ in the denominator matches $p(x|w^{(0)})$ and cancels out in the first term:

$$\left. \frac{\partial^2 K}{\partial w_j \partial w_k} \right|_{w=w^{(0)}} = -\int \frac{\partial^2 p(x|w^{(0)})}{\partial w_j \partial w_k} dx + \int p(x|w^{(0)}) \frac{\partial \log p(x|w^{(0)})}{\partial w_j} \frac{\partial \log p(x|w^{(0)})}{\partial w_k} dx$$

Since the integral of a probability distribution is 1 ($\int p(x|w)dx = 1$), its derivatives with respect to the parameters are zero:

$$\int \frac{\partial^2 p(x|w)}{\partial w_j \partial w_k} dx = 0$$

Thus, the first term vanishes, leaving:

$$\left. \frac{\partial^2 K}{\partial w_j \partial w_k} \right|_{w=w^{(0)}} = \int p(x|w^{(0)}) \left(\frac{\partial \log p(x|w^{(0)})}{\partial w_j} \right) \left(\frac{\partial \log p(x|w^{(0)})}{\partial w_k} \right) dx$$

By the definition of the Fisher Information Matrix (FIM), $I_{j,k}(w^{(0)}) = \mathbb{E}_x \left[\frac{\partial \log p}{\partial w_j} \frac{\partial \log p}{\partial w_k} \right]$. Thus:

$$\left. \frac{\partial^2 K(w)}{\partial w_j \partial w_k} \right|_{w=w^{(0)}} = I_{j,k}(w^{(0)})$$

2. Positive Definite Matrices

A matrix A is positive definite if for all non-zero column vectors x, we have $x^T Ax > 0$. By definition, a real positive definite matrix is symmetric ($A^T = A$).

Let v be an eigenvector of A and λ be its corresponding eigenvalue. Then:

$$Av = \lambda v$$

Pre-multiplying by v^T:

$$v^T Av = v^T(\lambda v) = \lambda(v^T v) = \lambda\|v\|^2$$

Since A is positive definite, for a non-zero eigenvector v, $v^T Av > 0$. Therefore:

$$\lambda\|v\|^2 > 0 \implies \lambda > 0$$

Thus, the matrix A must have strictly positive eigenvalues.

3. Rank of a Matrix and Determinant

Rank is the maximum number of linearly independent column vectors in a matrix. For example, if a 3×3 matrix has vectors v_1, v_2, v_3 and $v_3 = av_1 + bv_2$, then the rank is < 3. If the rank of an $n \times n$ matrix is n, it is full rank.

If a matrix A is not full rank, at least one column (say v_n) can be expressed as a linear combination of the others:

$$v_n = \sum_{i=1}^{n-1} a_i v_i$$

By determinant properties, we can perform column transformations without changing the determinant. If we apply the transformation:

$$C_n \leftarrow C_n - \sum_{i=1}^{n-1} a_i v_i$$

The last column becomes exactly zero. Any matrix with a zero column has a determinant of 0. Thus, for any non-full rank matrix, $|A| = 0$. Since $I(w^{(0)})$ being degenerate implies $\text{Rank}(I(w^{(0)})) < d$, it follows that $|I(w^{(0)})| = 0$.

4. Degeneracy in a Regression Model

For a regression problem with inputs x and targets y, we use the conditional log-likelihood with Gaussian noise:

$$\log p(y|x,w) = -\frac{1}{2}(y - f(x,w))^2 - \frac{1}{2}\log(2\pi)$$

Taking the partial derivative:

$$\frac{\partial \log p}{\partial w_j} = (y - f(x,w))\frac{\partial f(x,w)}{\partial w_j}$$

The FIM element $I_{j,k}(w^{(0)})$ is the expectation over the joint distribution $p(x,y|w^{(0)})$:

$$I_{j,k}(w^{(0)}) = \iint p(x,y|w^{(0)}) \left[(y - f(x,w))^2 \frac{\partial f}{\partial w_j}\frac{\partial f}{\partial w_k}\right] dydx$$

Since the variance of the Gaussian noise is assumed to be 1, the expectation over y is $\int p(y|x,w)(y - f(x,w))^2 dy = 1$.

The integral thus simplifies to:

$$I_{j,k}(w^{(0)}) = \int p(x)\frac{\partial f}{\partial w_j}\frac{\partial f}{\partial w_k}dx$$

We need to prove both ways since "iff" is mentioned in the problem

Forward Proof ($|I| = 0 \implies$ linearly dependent):
If $|I| = 0$, the matrix has a null space. Thus, $v^T I v = 0$ for some non-zero vector v.

$$v^T I v = \sum_{j=1}^{d}\sum_{k=1}^{d} v_j v_k I_{j,k} = \int p(x)\left(\sum_{j=1}^{d} v_j \frac{\partial f}{\partial w_j}\right)^2 dx = 0$$

Since $p(x) > 0$ and the squared term is always non-negative, the integral can only be zero if the inner term is zero everywhere:

$$\sum_{j=1}^{d} v_j \frac{\partial f(x,w)}{\partial w_j} = 0$$

Because v is non-zero, the set of derivatives is linearly dependent.

Reverse Proof (Linearly dependent $\implies |I| = 0$):
If the derivatives are linearly dependent, there exists a non-zero v such that $\sum_{j=1}^{d} v_j \frac{\partial f}{\partial w_j} = 0$. Squaring this, multiplying by $p(x)$, and integrating over x still yields 0. Reversing the summation steps gives $v^T I v = 0$, meaning , the matrix is skew symmetric , and therefore $|I(w^{(0)})| = 0$ (degenerate).

5. Singularity in a Two-Layer ReLU Network

If we take a purely linear model $f(x,w) = w_1 x_1 + w_2 x_2$, the derivatives $\frac{\partial f}{\partial w_1} = x_1$ and $\frac{\partial f}{\partial w_2} = x_2$ are linearly independent. The FIM is not degenerate, so it is NOT a singular model. We bring in a non-linearity $\mathrm{ReLU}(z) = \max(0,z)$ to introduce singularity.

Let us define a 2-layer neural network with 1 input, 1 hidden node, and 1 output:

$$f(x, \theta) = w_2 \text{ReLU}(w_1 x + b_1) + b_2$$

The parameter vector is $\theta = \{w_1, b_1, w_2, b_2\}$. We evaluate the partial derivatives. Let Q be the indicator function where $Q = 1$ if $(w_1 x + b_1) > 0$ and $Q = 0$ otherwise.

$$\begin{aligned}
\frac{\partial f}{\partial w_1} &= w_2 \cdot x \cdot Q \\
\frac{\partial f}{\partial w_2} &= \text{ReLU}(w_1 x + b_1) \\
\frac{\partial f}{\partial b_1} &= w_2 \cdot 1 \cdot Q \\
\frac{\partial f}{\partial b_2} &= 1
\end{aligned}$$

We can purposely set the true parameter $w_2 = 0$. If $w_2 = 0$, then regardless of the input x:

$$\frac{\partial f}{\partial w_1} = 0, \quad \frac{\partial f}{\partial b_1} = 0$$

To check for linear dependence, we set up the linear combination:

$$C_1 \frac{\partial f}{\partial w_1} + C_2 \frac{\partial f}{\partial w_2} + C_3 \frac{\partial f}{\partial b_1} + C_4 \frac{\partial f}{\partial b_2} = 0$$

Substituting the evaluated derivatives at $w_2 = 0$:

$$C_1(0) + C_2(\text{ReLU}(w_1 x + b_1)) + C_3(0) + C_4(1) = 0$$

We can satisfy this equation for all x by choosing $C_2 = C_4 = 0$ and non-zero constants $C_1 = 1, C_3 = 1$. Because a non-trivial combination equals zero, the set of derivatives is linearly dependent. Thus, the FIM is degenerate, proving this 2-layer ReLU neural network is a singular model.

§2 Indeed all Regular things are Normal

(Proof of Bernstein-Von Mises Theorem)

Posterior Equation: By Bayes' theorem, the log posterior is:

$$\ln(p(w|D_n)) = \ln(p(D_n|w)) + \ln(p(w)) + \text{Constant}$$

Second Order Taylor Expansion in Several Variables:

$$T(x) = f(a) + (x - a)^T \nabla f(a) + \frac{1}{2!}(x - a)^T \nabla^2 f(a)(x - a)$$

Apply this to $\ln(p(D_n|w))$ at $a = \hat{w}_n$ (Maximum Likelihood Estimator):

$$\ln(p(D_n|w)) \approx \ln(p(D_n|\hat{w}_n)) + (w - \hat{w}_n)^T \nabla \ln(p(D_n|\hat{w}_n)) + \frac{1}{2!}(w - \hat{w}_n)^T \nabla^2 \ln(p(D_n|\hat{w}_n))(w - \hat{w}_n)$$

Now: 1. Because $\hat{w}_n$ is the maximum, the first derivative is zero:

$$\nabla \ln(p(D_n|\hat{w}_n)) = 0$$

2. For the second derivative, we use the Fisher Information Matrix:

Proof:

$$I(w) = -\mathbb{E}_x[\nabla^2 \ln(p(x|w))] \approx -\frac{1}{n}\sum_{i=1}^{n} \nabla^2 \ln(p(x_i|w))$$

Multiply by n:

$$-nI(w) = \nabla^2 \ln p(D_n|w)$$

Evaluating at the MLE:

$$\nabla^2 \ln(p(D_n|\hat{w}_n)) = -nI(\hat{w}_n)$$

Substituting these back:

$$\ln(p(D_n|w)) \approx \ln(p(D_n|\hat{w}_n)) - \frac{n}{2}(w - \hat{w}_n)^T I(\hat{w}_n)(w - \hat{w}_n)$$

Now we Assume: $p(w) \approx p(\hat{w}_n)$
Reason: As $n \to \infty$, the likelihood function forms an incredibly sharp peak around the MLE. Because the posterior is restricted to this tiny neighborhood, the prior $p(w)$ is effectively flat/constant over this region.

Thus, substituting into our posterior equation:

$$\ln(p(w|D_n)) \approx \ln(p(D_n|\hat{w}_n)) - \frac{n}{2}(w - \hat{w}_n)^T I(\hat{w}_n)(w - \hat{w}_n) + \ln(p(\hat{w}_n)) + \text{const}$$

Since $\ln(p(D_n|\hat{w}_n))$ and $\ln(p(\hat{w}_n))$ are evaluated at a fixed point, they are just constants with respect to w. We absorb them into the constant term. Converting back from log to exponential:

$$p(w|D_n) \approx K \cdot \exp\left(-\frac{1}{2}(w - \hat{w}_n)^T (nI(\hat{w}_n))(w - \hat{w}_n)\right)$$

NOW: Multivariate Normal Distribution is

$$\mathcal{N}(\mu, \Sigma) = \frac{1}{(2\pi)^{d/2}} \frac{1}{\sqrt{|\Sigma|}} \cdot e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

By direct pattern matching ($\mu = \hat{w}_n$ and $\Sigma^{-1} = nI(\hat{w}_n) \implies \Sigma = \frac{1}{n}I(\hat{w}_n)^{-1}$):

$$p(w|D_n) \approx \mathcal{N}_d\left(\hat{w}_n, \frac{1}{n}I(\hat{w}_n)^{-1}\right)$$

As $n \to \infty$, by the consistency of the MLE, $\hat{w}_n \to w^{(0)}$ and $I(\hat{w}_n) \to I(w^{(0)})$. Therefore:

$$p(w|D_n) \to \mathcal{N}_d\left(w^{(0)}, \frac{1}{n}I(w^{(0)})^{-1}\right)$$

§3 Bayesian Information Criterion

Definition:

$$F_n = -\log(Z_n)$$

$$Z_n = \int \varphi(w) e^{-nL_n(w)} dw$$

Where $L_n(w)$ is the empirical negative log-likelihood.

We again apply the Second Order Taylor Expansion in several variables to $L_n(w)$ around the empirical minimizer $w^{(0)}$:

$$L_n(w) \approx L_n(w^{(0)}) + (w - w^{(0)})^T \nabla L_n(w^{(0)}) + \frac{1}{2}(w - w^{(0)})^T \nabla^2 L_n(w^{(0)})(w - w^{(0)})$$

Again:

$$\nabla L_n(w^{(0)}) = 0$$

$$\nabla^2 L_n(w^{(0)}) = I(w^{(0)})$$

(Note: FIM is positive here because L_n is the NEGATIVE log-likelihood)

$$\implies L_n(w) \approx L_n(w^{(0)}) + \frac{1}{2}(w - w^{(0)})^T I(w^{(0)})(w - w^{(0)})$$

Substitute this back into the integral for Z_n:

$$Z_n \approx \int \varphi(w) \exp\left(-nL_n(w^{(0)}) - \frac{n}{2}(w - w^{(0)})^T I(w^{(0)})(w - w^{(0)})\right) dw$$

$$Z_n \approx e^{-nL_n(w^{(0)})} \int \varphi(w) \exp\left(-\frac{n}{2}(w - w^{(0)})^T I(w^{(0)})(w - w^{(0)})\right) dw$$

Again, we assume $\varphi(w) \approx \varphi(w^{(0)})$
Reason: For large n, the term $e^{-\frac{n}{2}(\dots)}$ acts like a tight Gaussian weighting function. It heavily penalizes any w far from $w^{(0)}$, effectively shrinking the integration volume to a tiny neighborhood where the prior is roughly constant.

Pulling the prior out of the integral:

$$Z_n \approx e^{-nL_n(w^{(0)})} \varphi(w^{(0)}) \int \exp\left(-\frac{1}{2}(w - w^{(0)})^T (nI(w^{(0)}))(w - w^{(0)})\right) dw$$

NOW: d-dimensional Multivariate Gaussian

$$\int \exp\left(-\frac{1}{2}x^T A x\right) dx = \sqrt{\frac{(2\pi)^d}{|A|}}$$

Using $A = nI(w^{(0)})$:

$$Z_n \approx e^{-nL_n(w^{(0)})}\varphi(w^{(0)})\frac{(2\pi)^{d/2}}{|nI(w^{(0)})|^{1/2}}$$

Take the negative logarithm to find the free energy F_n:

$$-\log Z_n \approx nL_n(w^{(0)}) - \log(\varphi(w^{(0)})) - \frac{d}{2}\log(2\pi) + \frac{1}{2}\log|nI(w^{(0)})|$$

Using the property of determinants for a $d \times d$ matrix, $|nI| = n^d|I|$. Therefore $\log|nI| = d\log n + \log|I|$.

$$F_n \approx nL_n(w^{(0)}) + \frac{d}{2}\log n + \underbrace{\frac{1}{2}\log|I(w^{(0)})| - \frac{d}{2}\log(2\pi) - \log(\varphi(w^{(0)}))}_{\text{Constants } O(1)}$$

As $n \to \infty$, the first two terms ($O(n)$ and $O(\log n)$) completely dominate the constant terms. So:

$$\boxed{F_n \approx nL_n(w^{(0)}) + \tfrac{d}{2}\log n = \text{BIC}}$$

§4 Free parameters

Question: Understand what is meant by "minimally singular case". What is meant by "after some change of basis"? Take an example and write the steps of this change of basis. *Hint: Think of eigen values and vectors again.*

Definitions and Intuition

Singular matrix $[A_{n\times n}]$**:** A matrix where $\det(A) = 0$. This implies the columns are linearly dependent.

Minimally singular case: This occurs when the $\text{Rank}(A) = n - 1$. In this scenario, the matrix is "just barely" singular, having exactly one dimension that collapses.

Change of Basis: In the standard Cartesian coordinate space, $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}0\\1\end{bmatrix}$ are the basis vectors. Any vector in 2-D space can be expressed as a linear combination of these basis vectors.

Changing the vector implies that these basis vectors are changed, but they can still describe the 2-D space (basically a change of coordinate axes). We can express the new basis vectors as a matrix multiplication of the original basis vectors since the original ones anyway span the whole space.

The Singularity Mapping: Now, say $Av = \lambda v$. If $\lambda = 0$, then $Av = 0$. A **singularity** $\implies$ there is a vector that gets mapped to the zero vector upon transformation. Because of this, there is no way to take that zero vector and get back to the original vector; therefore, **no inverse exists**.

Information is lost when doing the transformation. It is like collapsing a dimension:

- Cube $\to$ Sheet
- Sheet $\to$ Line

For a "minimally singular" matrix, it has exactly one zero eigenvalue. We will get exactly one free parameter. By changing the basis to the matrix's eigenvectors, we get a diagonal matrix (other terms become 0), which helps to find the free parameter.

Step-by-Step Example: Change of Basis

Say $Ax = 0$ with $A = \begin{bmatrix}2 & -2\\-2 & 2\end{bmatrix}$.

1. Find the Eigenvalues (λ):

$$\det(A - \lambda I) = 0 \implies (2-\lambda)(2-\lambda) - (-2)(-2) = 0$$

$$\lambda^2 - 4\lambda + 4 - 4 = 0 \implies \lambda(\lambda - 4) = 0$$

$$\lambda_1 = 0, \quad \lambda_2 = 4$$

2. Find Eigenvectors (v): For $\lambda_1 = 0$:

$$\begin{bmatrix}2 & -2\\-2 & 2\end{bmatrix}\begin{bmatrix}n_1\\n_2\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \implies 2n_1 - 2n_2 = 0 \implies n_1 = n_2 \implies v_1 = \begin{bmatrix}1\\1\end{bmatrix}$$

For $\lambda_2 = 4$:

$$\begin{bmatrix}-2 & -2\\-2 & -2\end{bmatrix}\begin{bmatrix}n_1\\n_2\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \implies n_1 + n_2 = 0 \implies v_2 = \begin{bmatrix}1\\-1\end{bmatrix}$$

3. Transformation Matrix (P): P consists of columns as eigenvectors: $P = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$. Let $x = Py$. Substituting: $Ax = 0 \implies A(Py) = 0$. Multiplying by P^{-1} on both sides: $P^{-1}(APy) = 0$. This $P^{-1}AP$ becomes a diagonal matrix of eigenvalues.

Proof of Diagonalization: $AP = (Av_1, Av_2) = (\lambda_1 v_1, \lambda_2 v_2)$. Let $B = \begin{bmatrix}\lambda_1 & 0\\0 & \lambda_2\end{bmatrix}$. Then $PB = (v_1, v_2)\begin{bmatrix}\lambda_1 & 0\\0 & \lambda_2\end{bmatrix} = (\lambda_1 v_1, \lambda_2 v_2)$. So, $AP = PB \implies P^{-1}AP = B$ (diagonal matrix).

4. Identifying the Free Variable: Our system is now:

$$\begin{bmatrix}0 & 0\\0 & 4\end{bmatrix}\begin{bmatrix}y_1\\y_2\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$$

$0 \cdot y_1 = 0 \implies y_1$ is a **free variable**. $4 \cdot y_2 = 0 \implies y_2 = 0$.

Since $x = Py$:

$$x = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ 0 \end{bmatrix} = y_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

So, x depends on y_1. All the infinite solutions depend on 1 free variable, i.e., y_1.

§5 Get the feel

Question: Cook some examples of Singularities which are free parameters and which are not. Atleast two each. Also obviously show the math of the Hessian/ FIM.
Get the K v/s w plots for each of these. I want to see interesting shapes :)

Singularity vs. Free Parameter

Singularity: Geometry of the loss landscape.
Free parameter: Feature of the model's architecture.

Singularity: A specific point in your parameter space where the curvature of the loss function completely flattens in ≥ 1 directions. For this, you should have the Hessian matrix (determinant) $= 0$ (at least one eigenvalue $= 0$). Meaning if you move in that direction, it doesn't change the slope or the curvature.

Free Parameter: A variable that is entirely redundant. If you move in this parameter space, the model's output or loss don't change. So, **every free parameter creates a singularity**.

Singularities that are free parameters: The 0 eigenvalue corresponds to a direction in parameter space where the loss function is completely independent.

Examples: Singularities that ARE Free Parameters

1. Additive Redundancy

When multiple parameters do essentially the same job (where you have way too many parameters for no reason). For example: $f(x) = (w_1 + w_2)x$, if the target is $f(x) = 0$.

$$Loss : K = \frac{1}{2}(w_1 + w_2)^2$$

$$H = \begin{bmatrix} \frac{\partial^2 K}{\partial w_1^2} & \frac{\partial^2 K}{\partial w_1 \partial w_2} \\ \frac{\partial^2 K}{\partial w_1 \partial w_2} & \frac{\partial^2 K}{\partial w_2^2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Singularities of H are $\lambda_1 = 2$ and $\lambda_2 = 0$, with eigenvector $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. If you move along $w_1 + w_2 = 0$, the loss is $K = 0$ forever.

2. Multiplicative Redundancy

Say $f(x) = (w_1 w_2)x$. $K(w_1, w_2) = \frac{1}{2}(w_1 w_2)^2$.

$$H = \begin{bmatrix} w_2^2 & 2w_1 w_2 \\ 2w_1 w_2 & w_1^2 \end{bmatrix}, \quad \text{At origin } H_{(0,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Since both eigenvalues are 0 at the origin, if you set $w_1 = 0$, then $K(0, w_2) = 0$, and it doesn't matter what the value of w_2 is.

Examples: Singularities that are NOT Free Parameters

1. Flat Bottomed Bowl

When a parameter's relationship to the loss is governed by a greater even power: $K(w) = \frac{1}{2}w^4$.

$$H = 6w^2$$

At minimum, $w = 0$, $H = 0$. So even though the eigenvalue is zero at the origin, you can't move it, else $H \neq 0$.

2. The Walled Trench

When one parameter is like a quadratic and the other is of higher power (for the Hessian to be non-constant). $K(w_1, w_2) = \frac{1}{2}w_1^2 + \frac{1}{2}w_2^4$.

$$H = \begin{bmatrix} 1 & 0 \\ 0 & 6w_2^2 \end{bmatrix}, \quad \text{At origin } H_{(0,0)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

At the origin, $\lambda_1 = 1, \lambda_2 = 0$. The graph is similar to the above case as the higher power starts to dominate.

§6 Learning Coefficient

Question: Get the plot of $V(\epsilon)$ for the examples above (for different epsilons set). What does this idea of taking volume benefit from compared to the earlier method. Try to be mathematically rigorous. When can the dimensionality be "read-off" from RLCT and when not?

Mathematical Rigor: Volume of Parameter Space

$V(\epsilon)$ measures the volume of the parameter space where the loss function $K(w)$ is less than ϵ. The $V(\epsilon)$ plots, as ϵ becomes tiny, scale according to the formula:

$$V(\epsilon) \propto \epsilon^{\lambda}(-\log \epsilon)^{m-1}$$

Where:

- λ: Real Log Canonical Threshold (RLCT)
- m: Multiplicity

Volume Scaling for Previous Examples

Additive ($K = \frac{1}{2}(w_1 + w_2)^2$): Condition: $|w_1 + w_2| \leq \sqrt{2\epsilon} \implies$ Volume $\propto \sqrt{\epsilon}$.

Multiplicative ($K = \frac{1}{2}w_1^2 w_2^2$): Condition: $|w_1 w_2| < \sqrt{2\epsilon}$. Area $= \int \frac{1}{w_1} dw_1 \implies$ Volume $\propto \sqrt{\epsilon}(-\log \epsilon)$.

Symmetrically Degenerate ($K = \frac{1}{2}w^{2n}$): Condition: $|w| < \sqrt[2n]{2\epsilon} \implies$ Volume $\propto \epsilon^{1/2n}$.

Anisotropic Degenerate ($K = \frac{1}{2}w_1^2 + \frac{1}{2}w_2^4$): Volume $\propto \epsilon^{1/2} \times \epsilon^{1/4} = \epsilon^{3/4}$.

Flaw of the Hessian

The **Hessian** is a second-order Taylor approximation, so the function is always assumed to be a quadratic bowl:

$$K(w) \approx \frac{1}{2}w^T H w$$

So, volume$(\epsilon) \propto \frac{\epsilon^{d/2}}{\sqrt{|\det H|}}$ (In d-dimensions, instead of power 3/2 it'll be $d/2$).

Another flaw is at the **singularity**, since the determinant of the Hessian is 0, the volume becomes ∞. In the new method (Singular Learning Theory), the higher-order terms are **not dropped**, allowing for a rigorous calculation of the volume scaling even at the singular points.